



EXPERIMENTAL AND COMPUTATIONAL HEAT TRANSFER IN THERMAL MANAGEMENT OF ELECTRONICS

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OUTLINE

- I. Overview of Heat Transfer Laboratory
- II. Current relevant projects at UA
 - A. Liquid Cooling
 - Innovative Boiling Enhanced Millichannel Heat Sinks
 - Gas Assisted Evaporative Cooling
 - Thermosyphons
 - B. Air Cooling
 - Pin Fin Heat Sinks in Cross Flow
 - Pin Fin Heat Sinks in Impinging flow
 - C. Compact Thermal Modeling
 - D. Rapid Thermo-fluid Prototyping
 - E. Characterization of Advanced Interfacial Materials



OVERVIEW

- ◆ Experimental and Computational Convective Heat Transfer: Fundamental and Applied
 - Electronics Cooling
 - Gas Turbine Heat Transfer
- ◆ Research Sponsors:
 - NSF, AFOSR, DOE, SRC, Raytheon/TI, Intel, AMD, Honeywell, Cisco, HP/Compaq
- ◆ Facilities:
 - Heat Transfer Laboratory
 - Computational Heat Transfer Laboratory
- ◆ People:
 - PhD (3), MS (10), BS (11)



FACILITIES

- ◆ Open Return Wind Tunnel, 0-20 m/s
- ◆ Closed Return Wind Tunnel, 0-25 m/s
- ◆ Impinging Jet Heat Transfer Facility
- ◆ Interfacial Heat Transfer Testing Apparatus
- ◆ Heat Sink Characterization Apparatus
- ◆ Rapid Thermal Prototyping Facility
- ◆ Numerous Benchtop Rigs



FACILITIES



Annular Wind Tunnel



Closed Return Wind Tunnel



Open Return Wind Tunnel



Linear Cascade Wind Tunnel



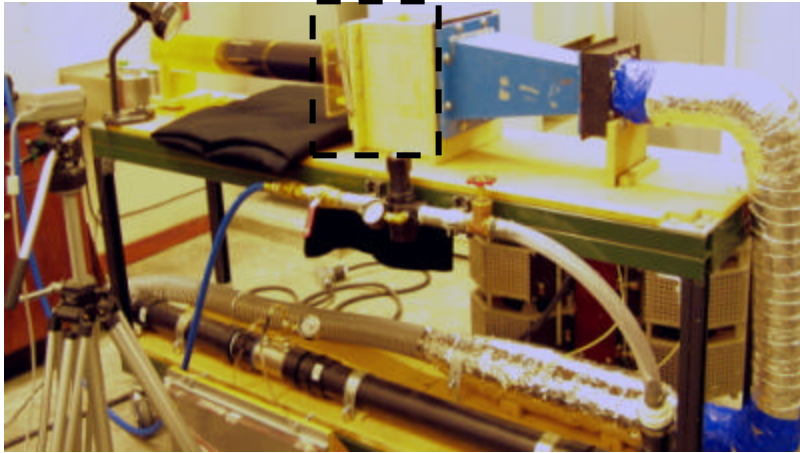
FACILITIES



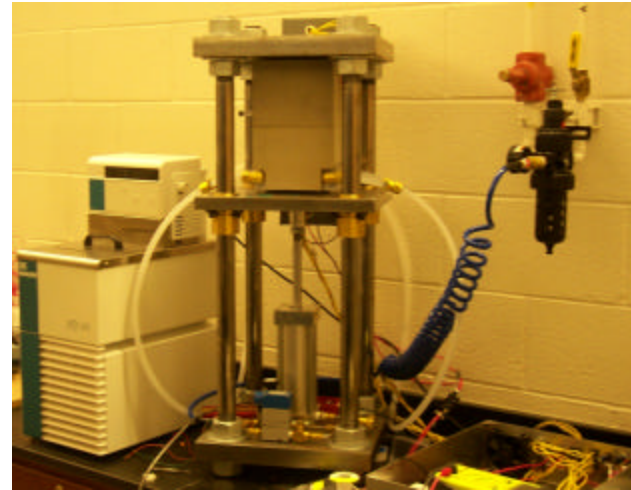
Jet Impingement Rig



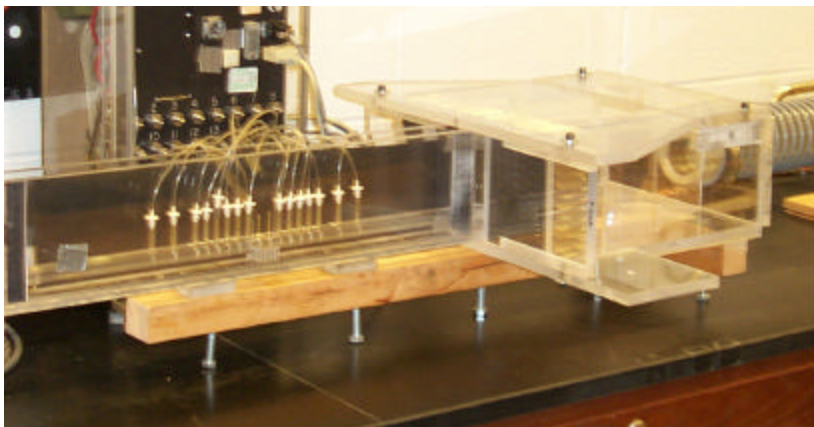
BENCHTOP RIGS



Transient Liquid Crystal Rig



Interfacial and Contact Resistance Rig

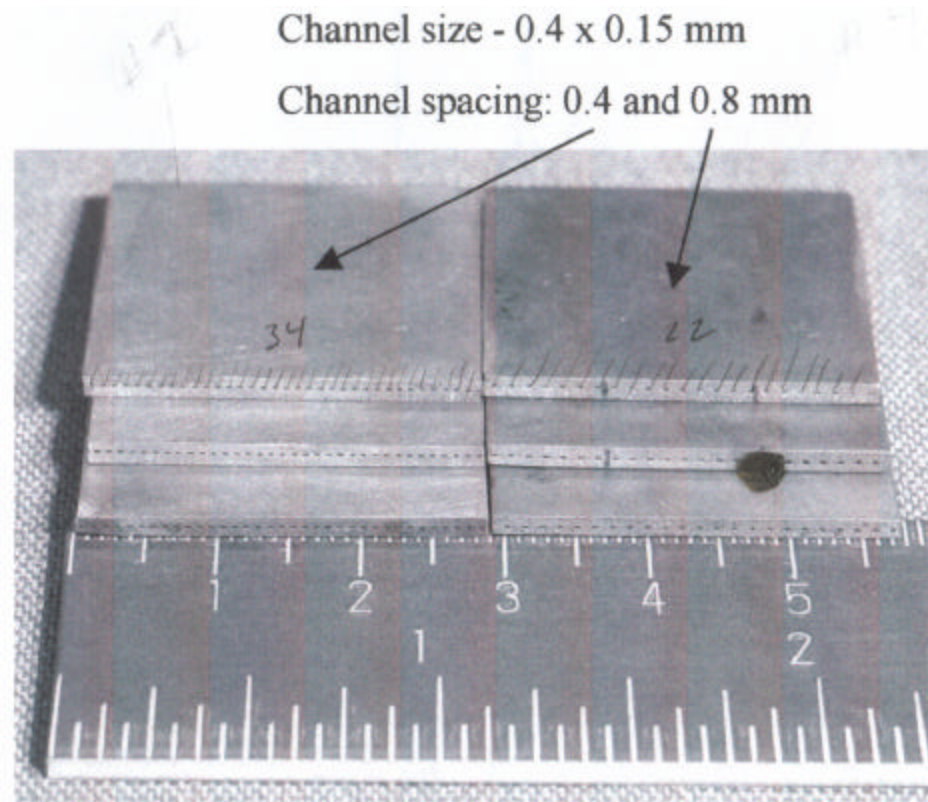


Heat Sink Characterization Rig



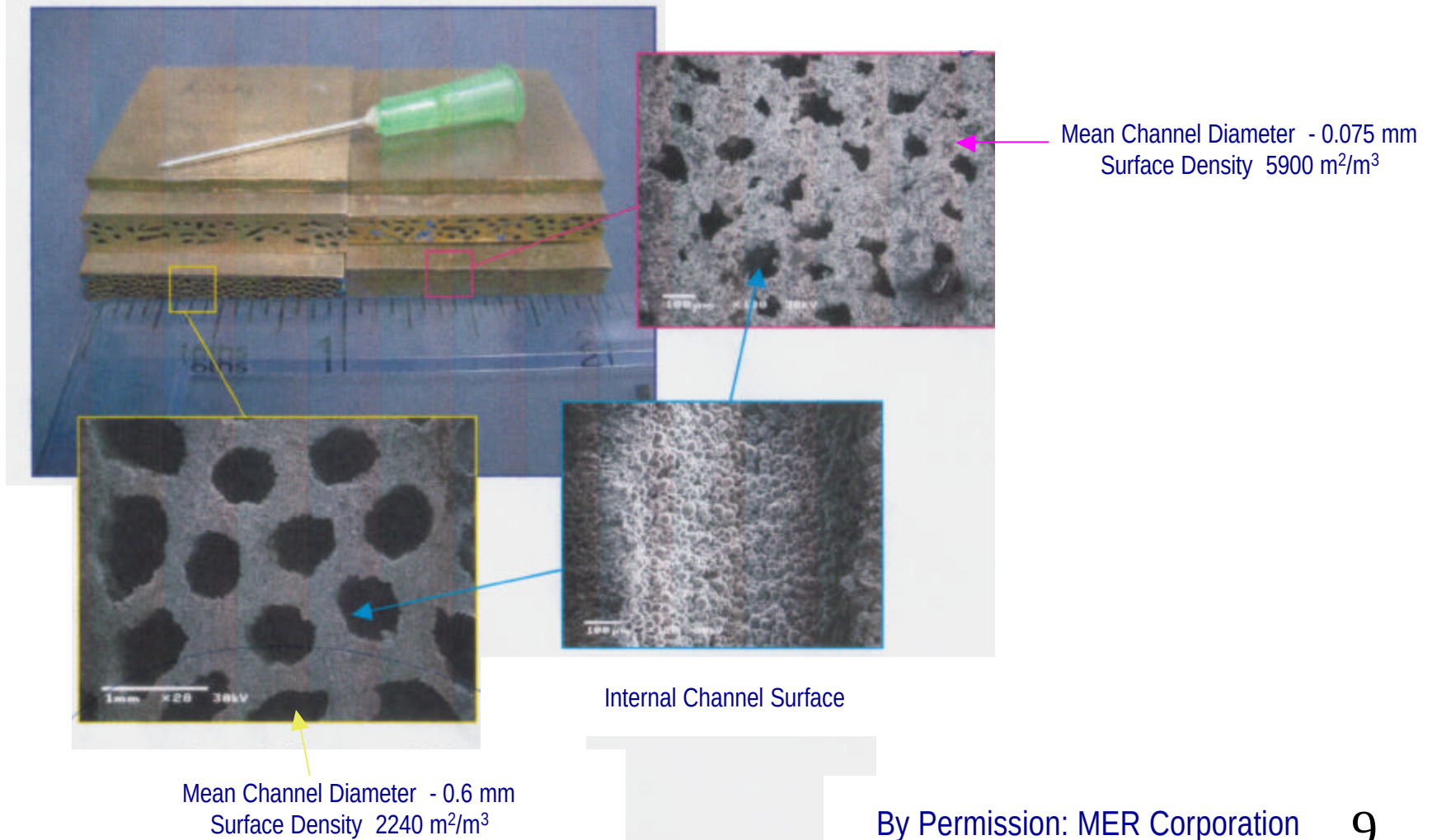
Novel Polycapillary Channel Liquid Cooled Heat Sinks

- ◆ Develop methods for manufacturing heat sinks with enhanced internal surfaces
- ◆ 500 – 1000 W/cm²
- ◆ Heat sinks manufactured using novel powder metallurgy (metal / ceramic)
- ◆ Capable of producing highly organized or random porous structures





Novel Polycapillary Channel Liquid Cooled Heat Sinks

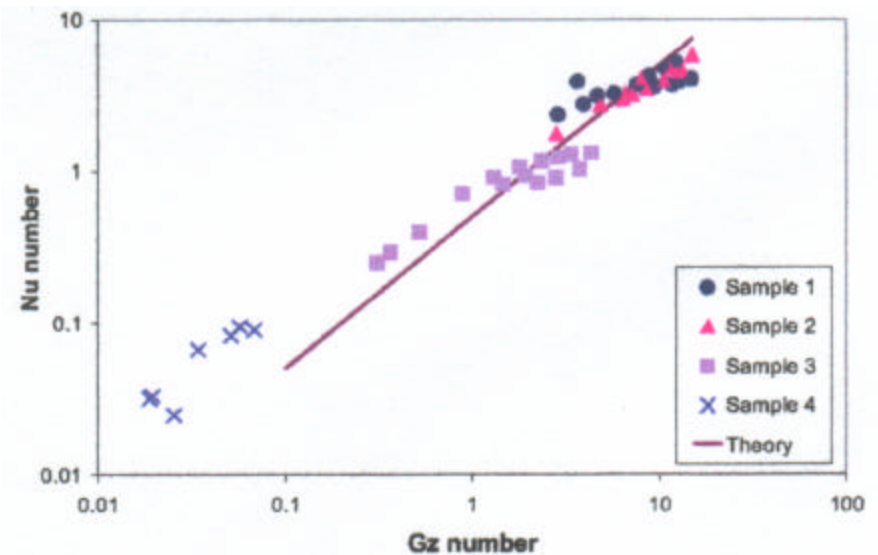
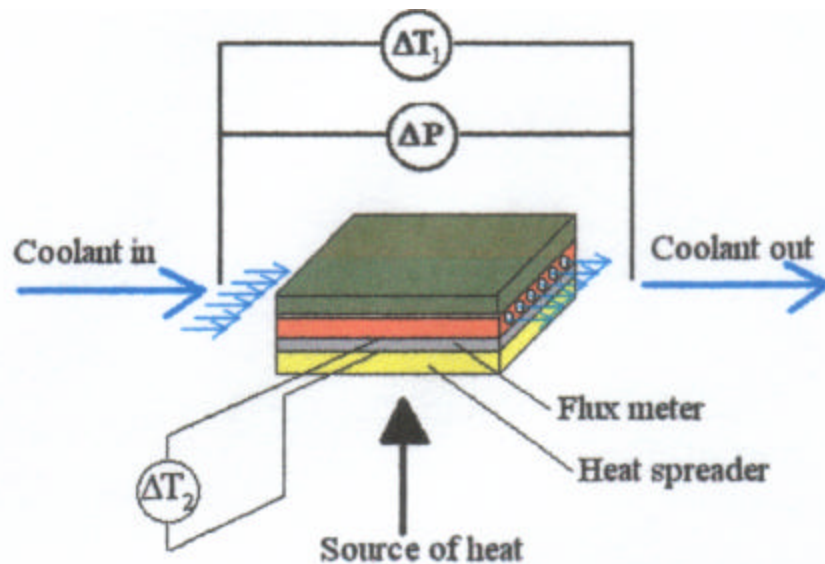


By Permission: MER Corporation



Novel Polycapillary Channel Liquid Cooled Heat Sinks

Typical Results in Single Phase flow



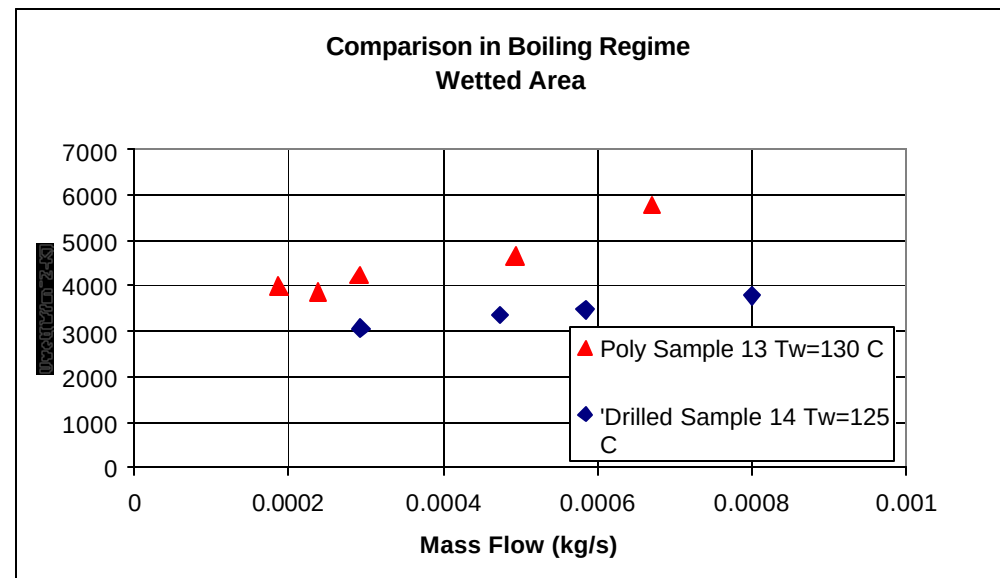


Novel Polycapillary Channel Liquid Cooled Heat Sinks

Enhancement of Boiling

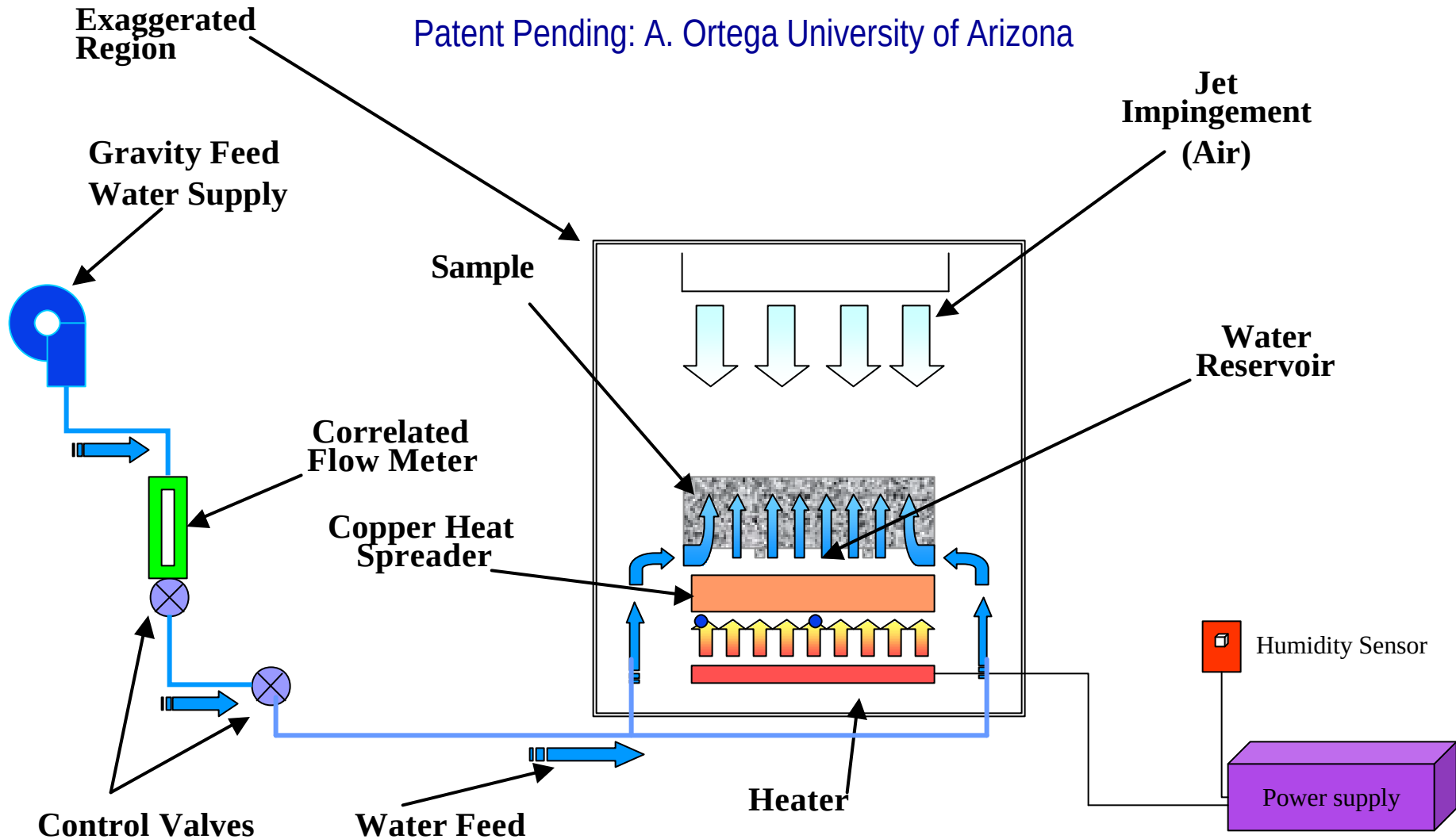
Comparison of Milled Channel and Polycapillary Channel

- ◆ Polycapillary channel produces apparent enhancement of boiling heat transfer
- ◆ Detailed mechanisms not understood
- ◆ Subject of continuing study



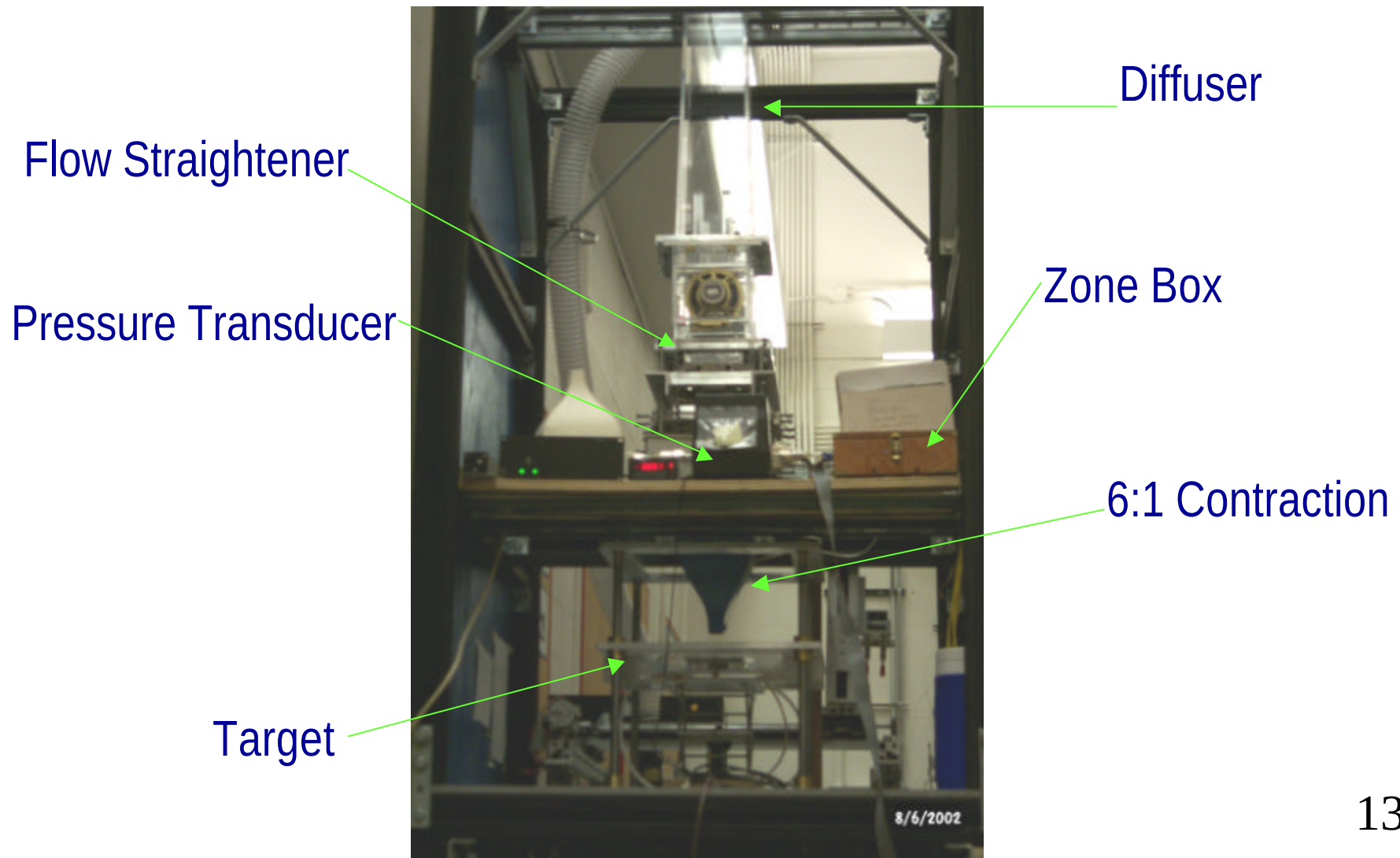


Gas Assisted Liquid Evaporative Cooling





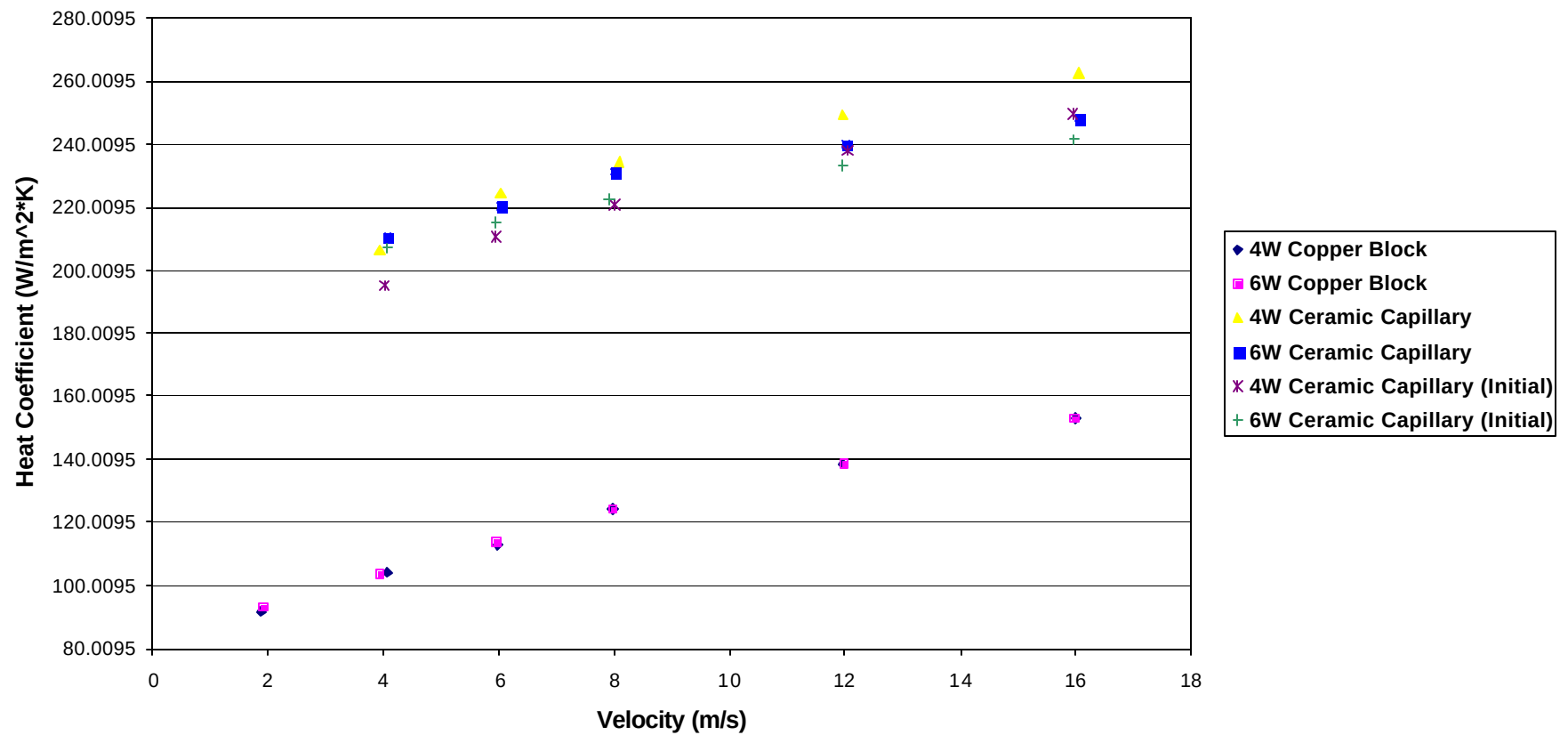
Gas Assisted Liquid Evaporative Cooling





Gas Assisted Liquid Evaporative Cooling

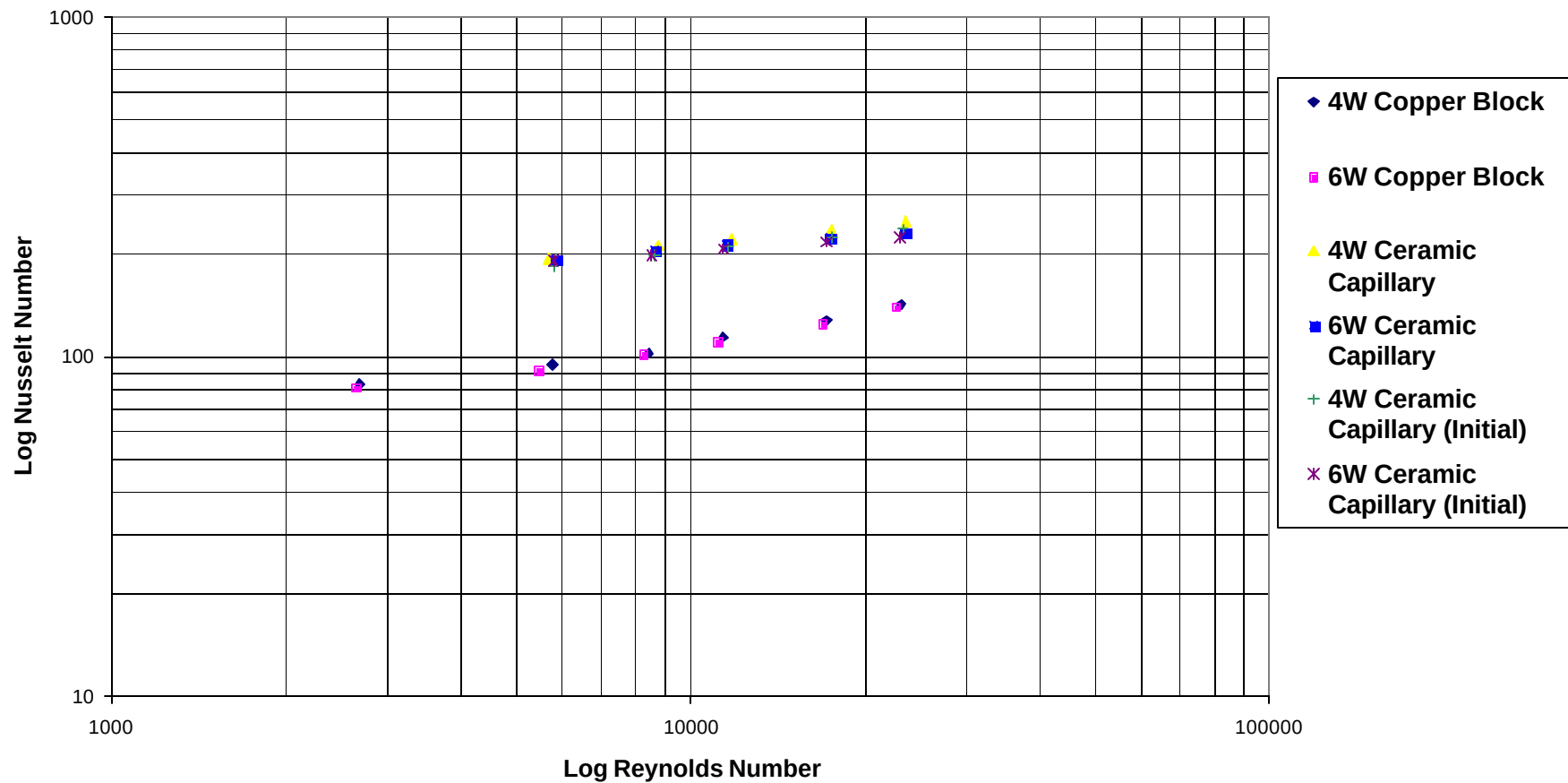
Heat Transfer Coefficient vs. Velocity





Gas Assisted Liquid Evaporative Cooling

Log Nusselt Number vs. Log Reynolds Number



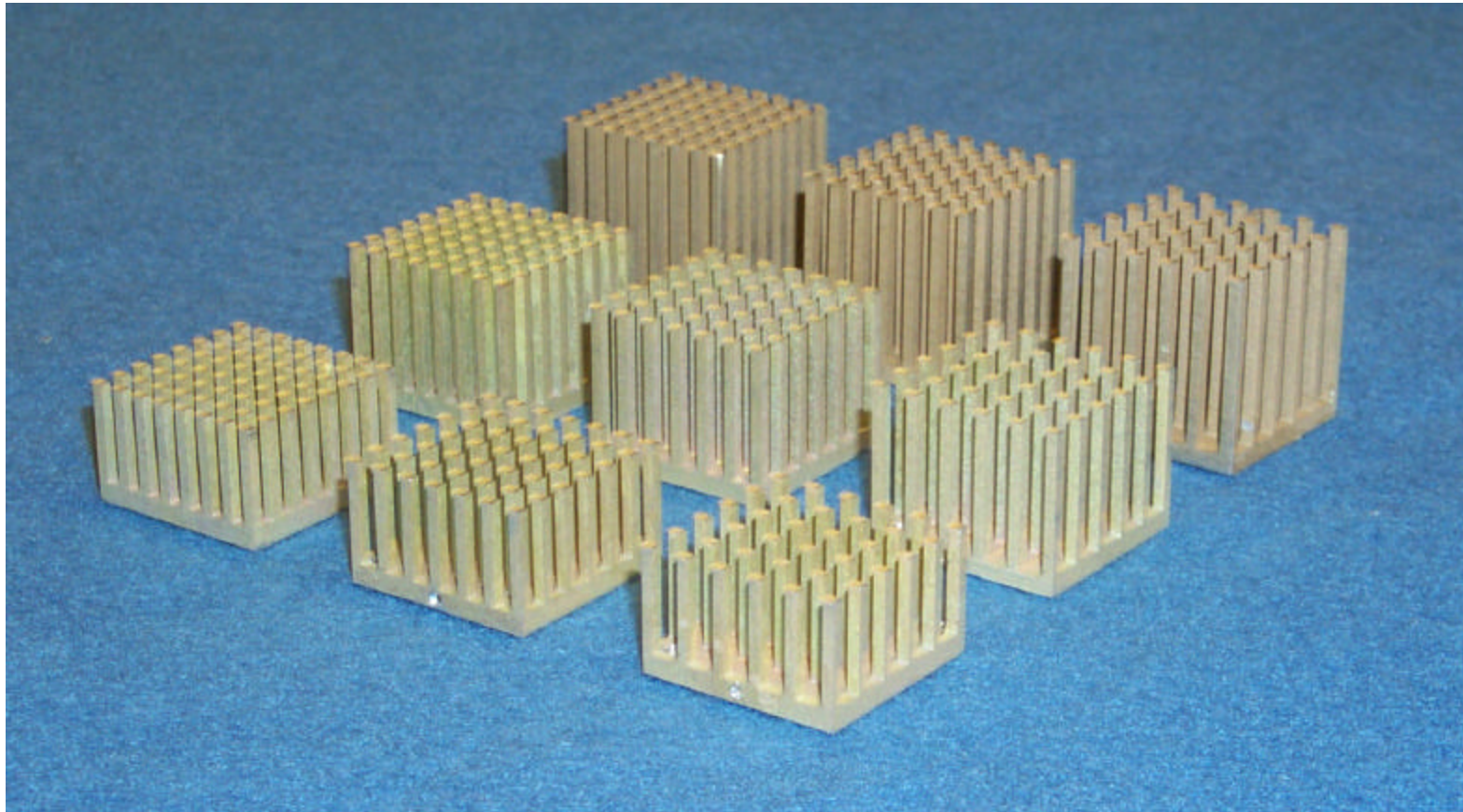


Heat Transfer and Fluid Mechanics in Pin Fin Heat Sinks

- ◆ Experimentally and Computationally investigate the flow and convective heat transfer in Pin Fin Heat Sinks
- ◆ In Side Inlet and Top Inlet configurations
- ◆ Extract
 - Inlet and Exit Loss Coefficients
 - Core Loss Coefficients
 - Heat Transfer Coefficients
- ◆ Develop design-compatible 1-Dimensional Modeling tool
- ◆ Embed tool in UATDMT



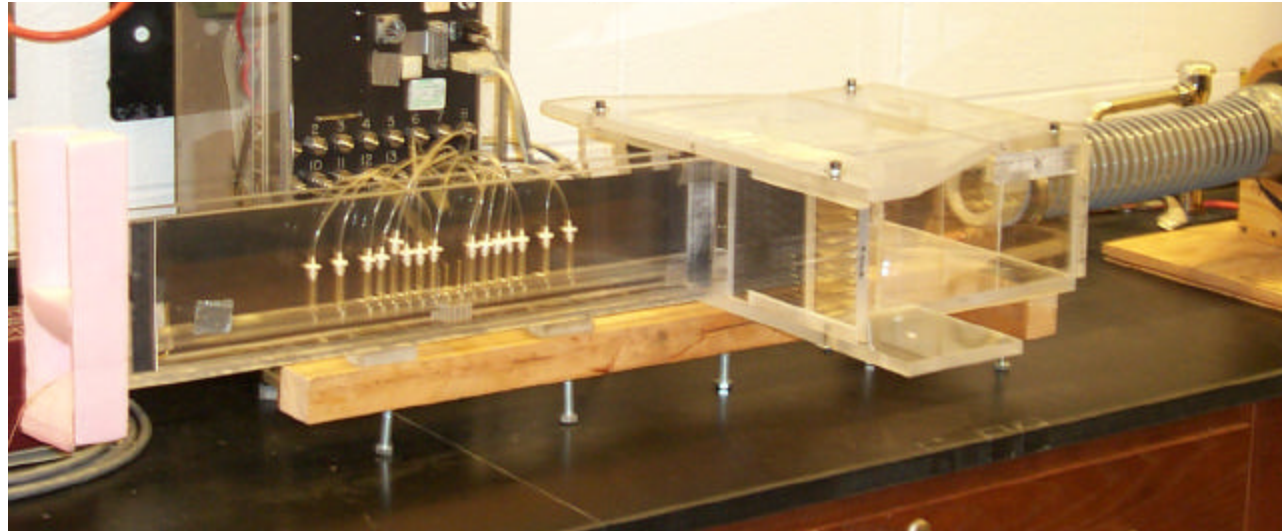
Heat Transfer and Fluid Mechanics in Pin Fin Heat Sinks



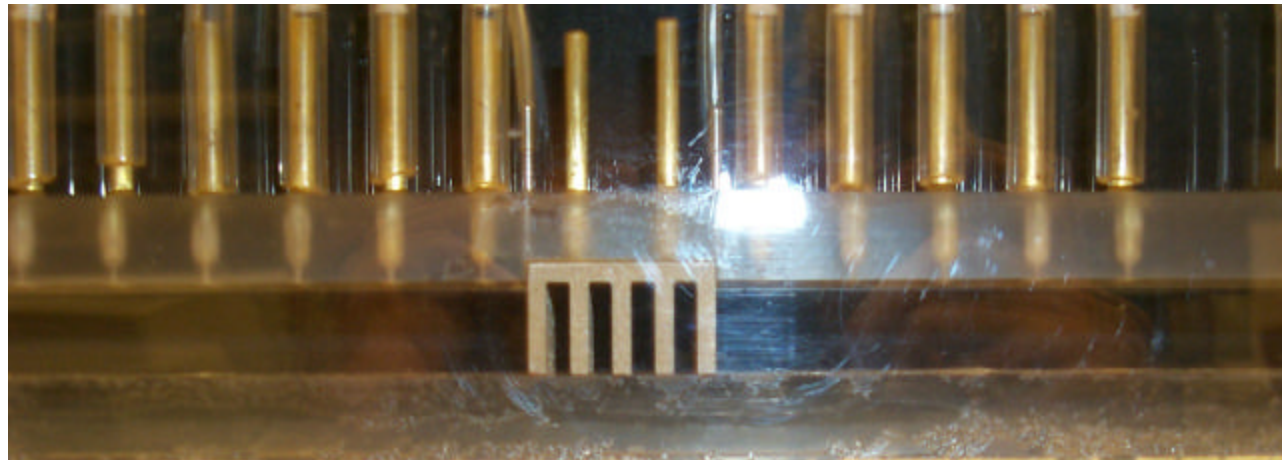


Experimental Investigation of PF Heat Sinks

Wind Tunnel



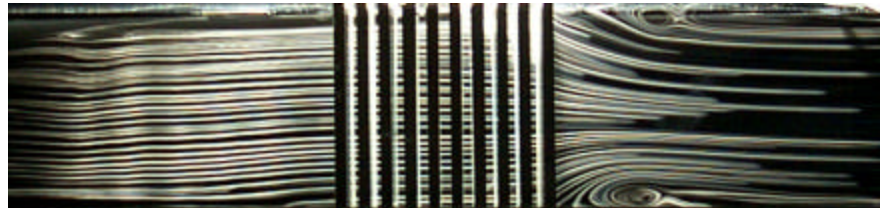
Close up showing pressure taps



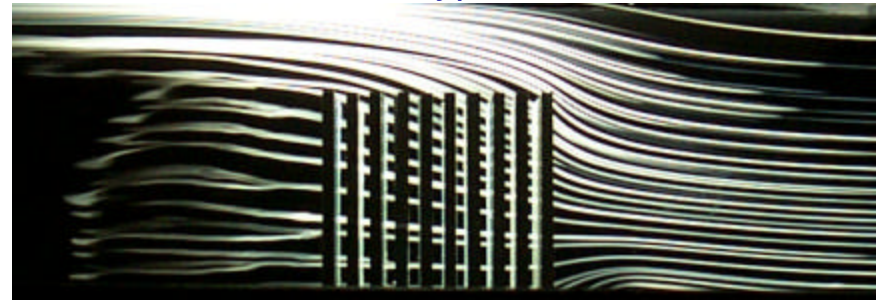


Behavior of SISE Pin Fin Heat Sinks

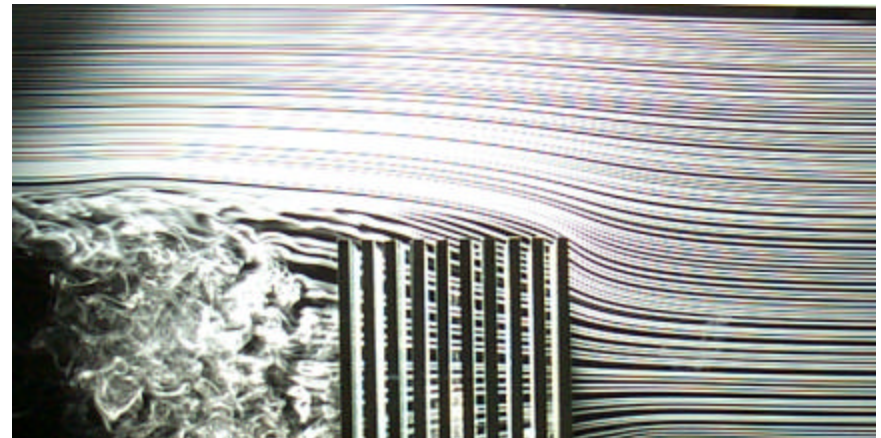
- ◆ Flow Bypass reduces the cooling flow
- ◆ Flow experiences transitional behavior that is dependent on approach Re and bypass ratio



$S/t=2$ Bypass=0



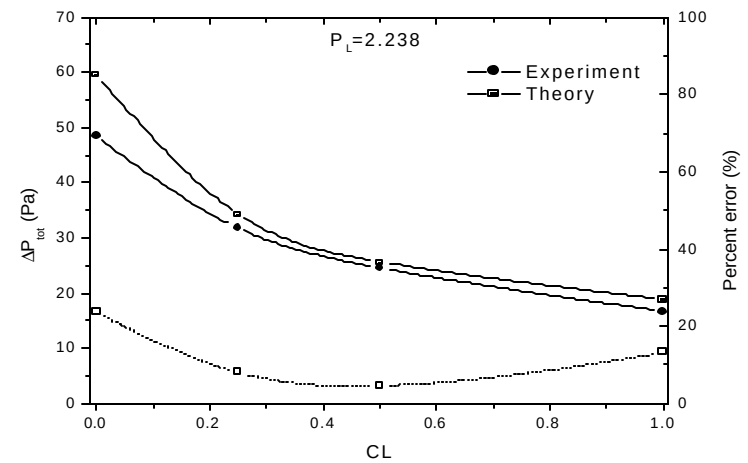
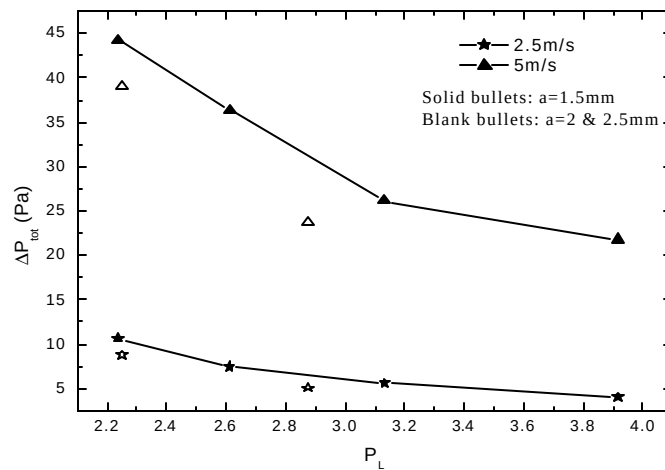
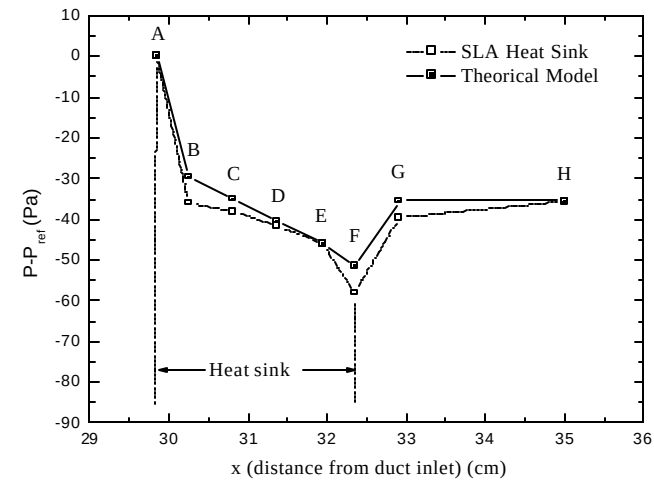
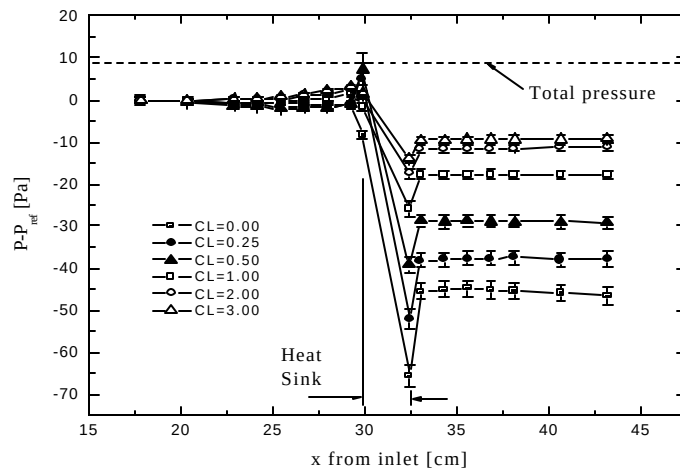
$S/t=2$ Bypass=1/2



$S/t=2$ Bypass=1



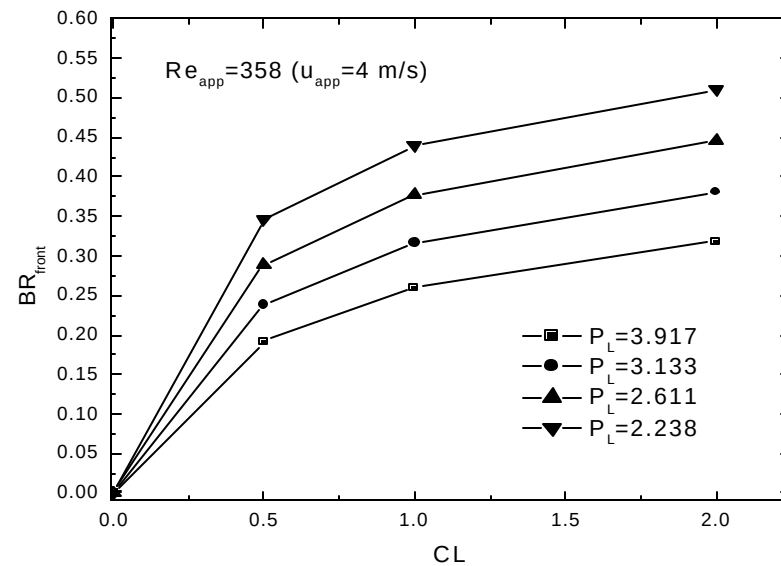
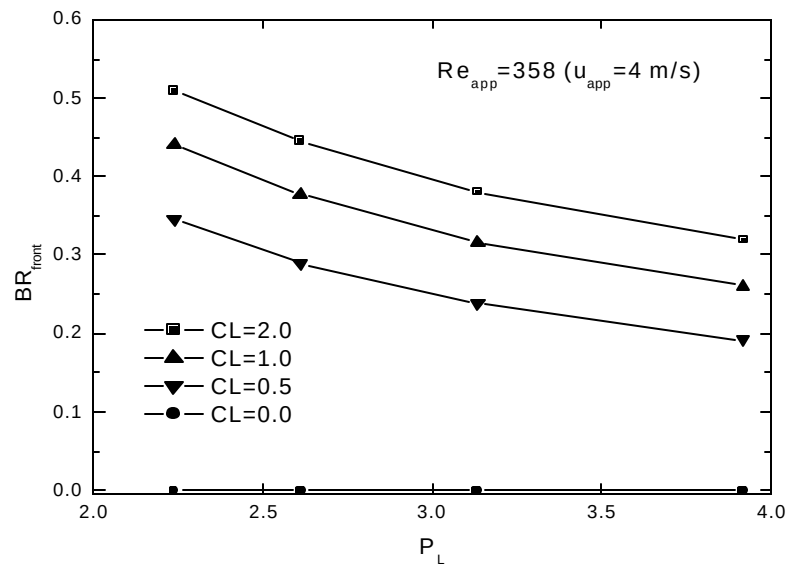
SISE Pin Fin Heat Sinks: Experimental Data





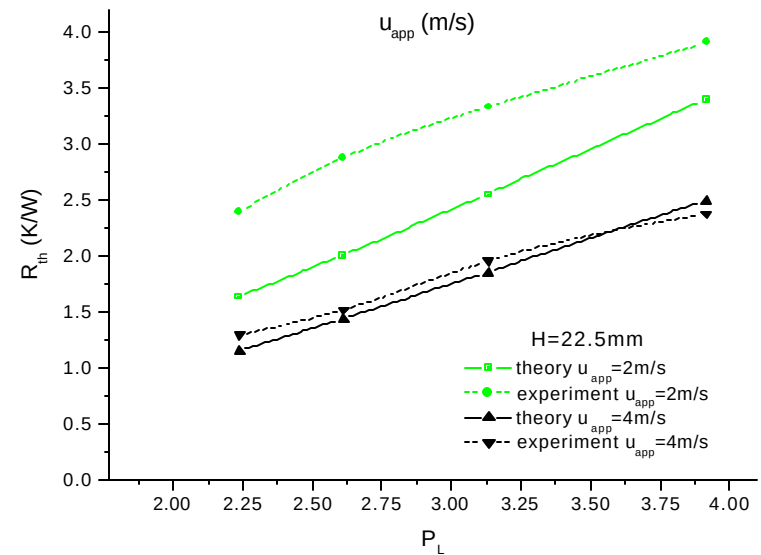
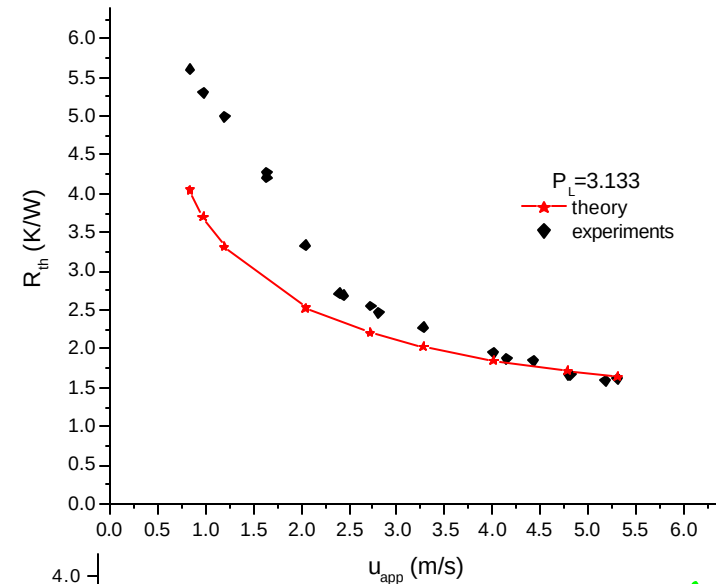
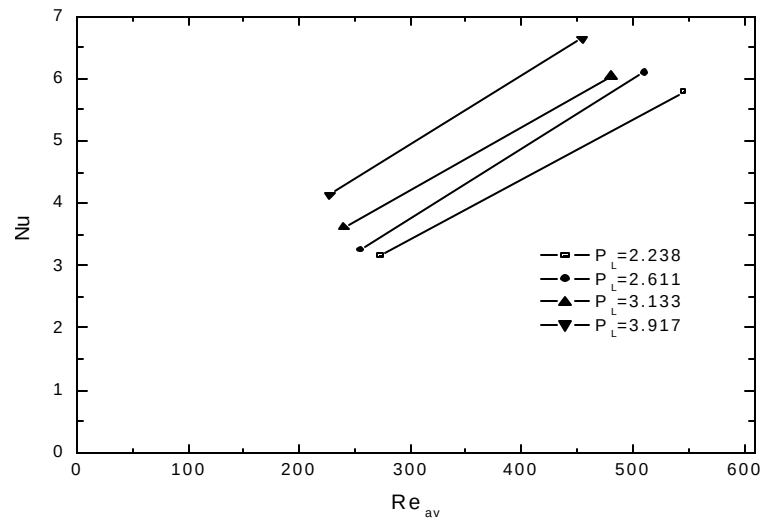
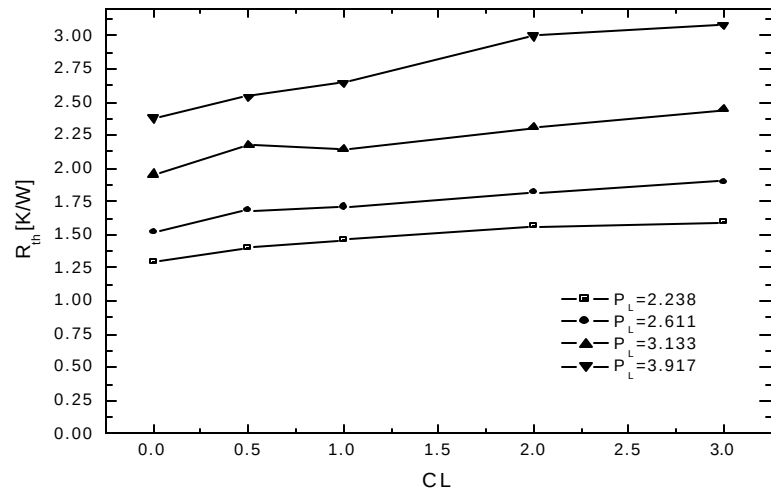
SISE Pin Fin Heat Sinks: Simulation Results

Predictions of Flow Bypass



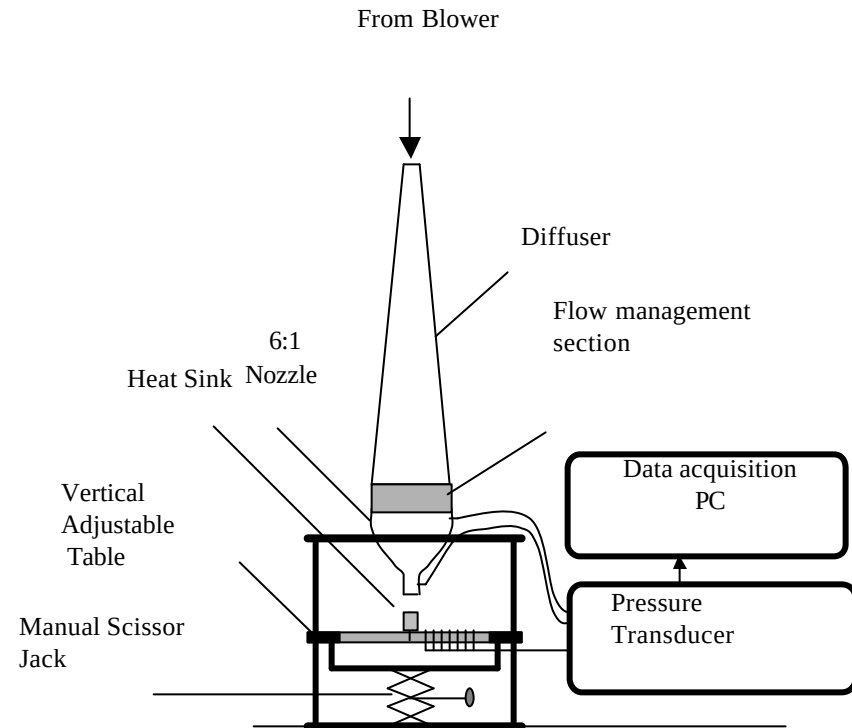
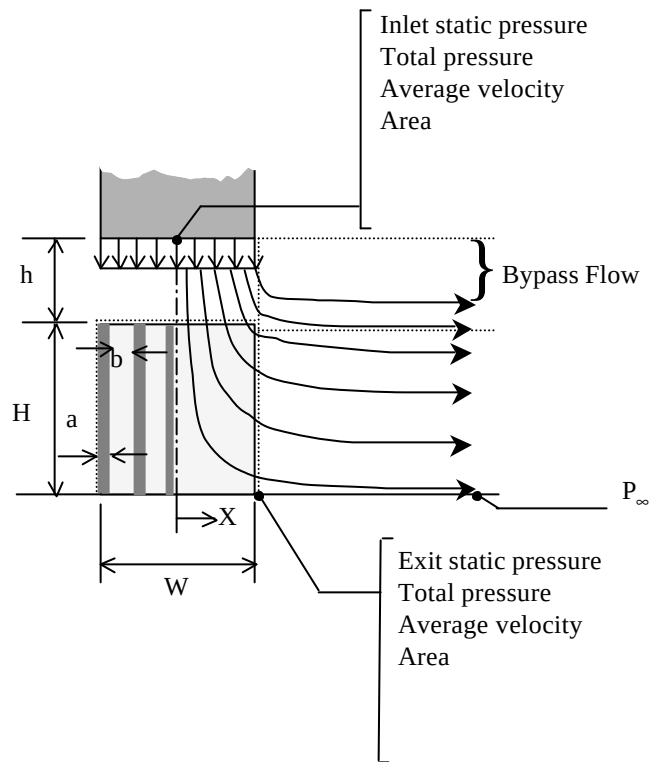


SISE Pin Fin Heat Sinks: Thermal Results



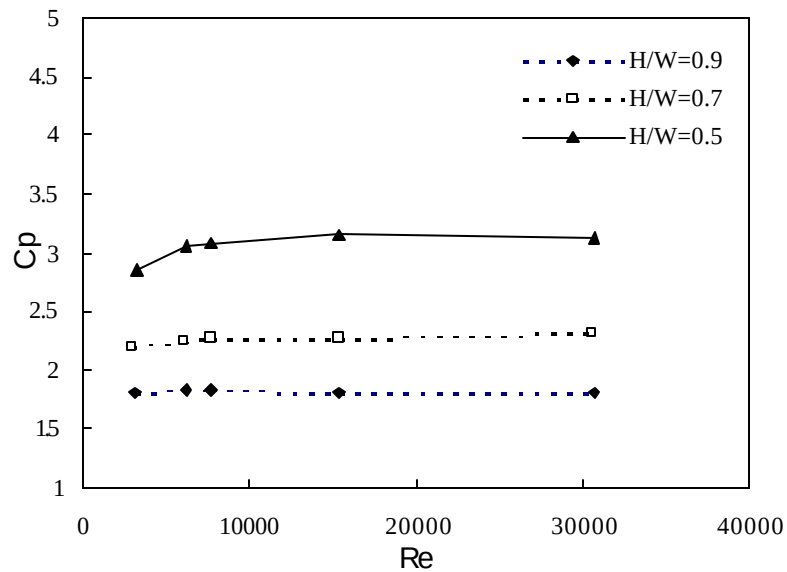


TISE Pin Fin Heat Sinks (Impingement)

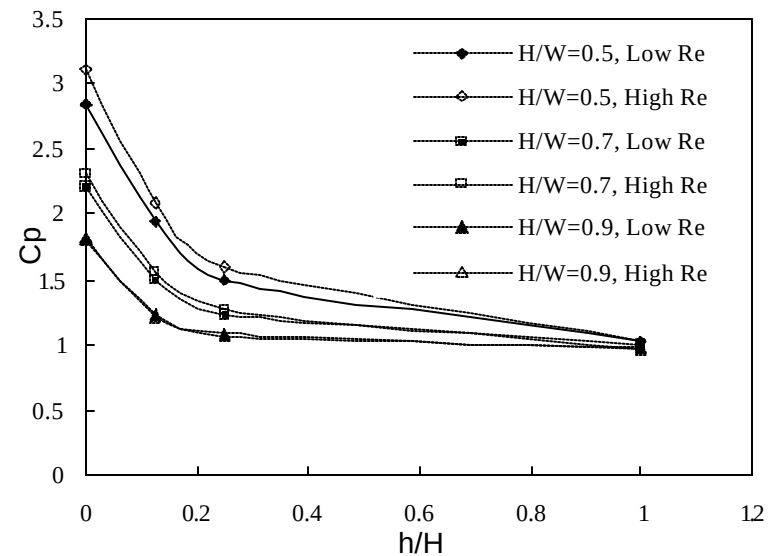




TISE Pin Fin Heat Sinks



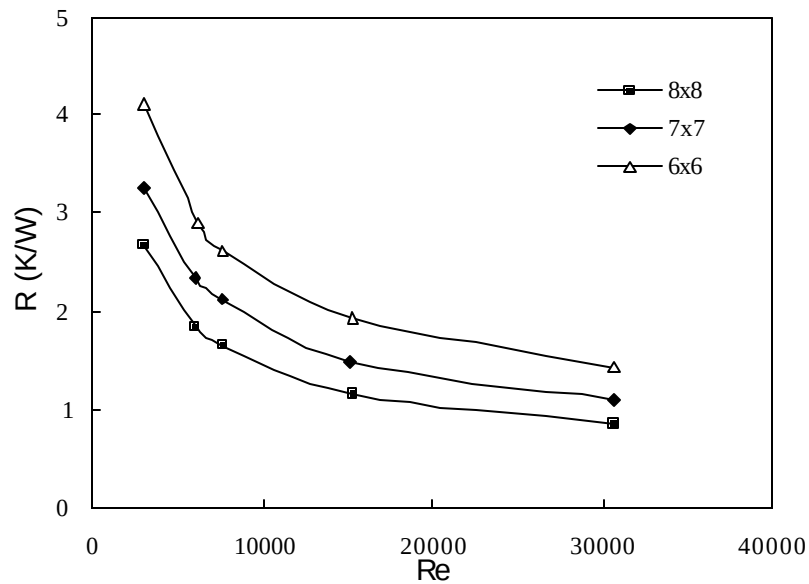
Dependence of pressure loss coefficient on Reynolds number for heat sinks of different heights 1A, 1B and 1C (8x8) at $h/H = 0$.



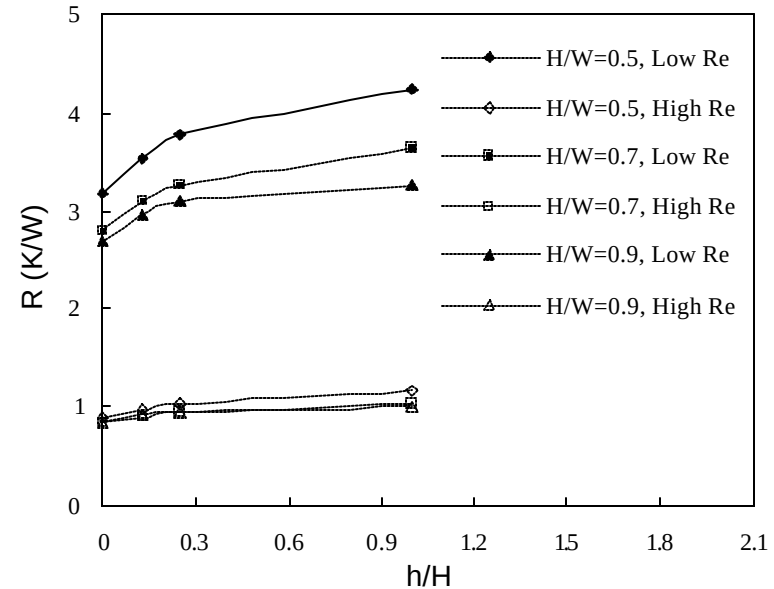
Effect of tip clearance on overall pressure loss coefficient for 1A, 1B and 1C (8x8) at various Re .



TISE Pin Fin Heat Sinks



Heat sink base-to-ambient resistance dependence on Reynolds number and pin density for $a/H = 0.12$ at fixed pin height $H/W = 0.9$ and $h/H = 0$.



Overall thermal resistance vs. clearance ratio at high Re (30600) and low Re (3000) for 8x8 heat sink.



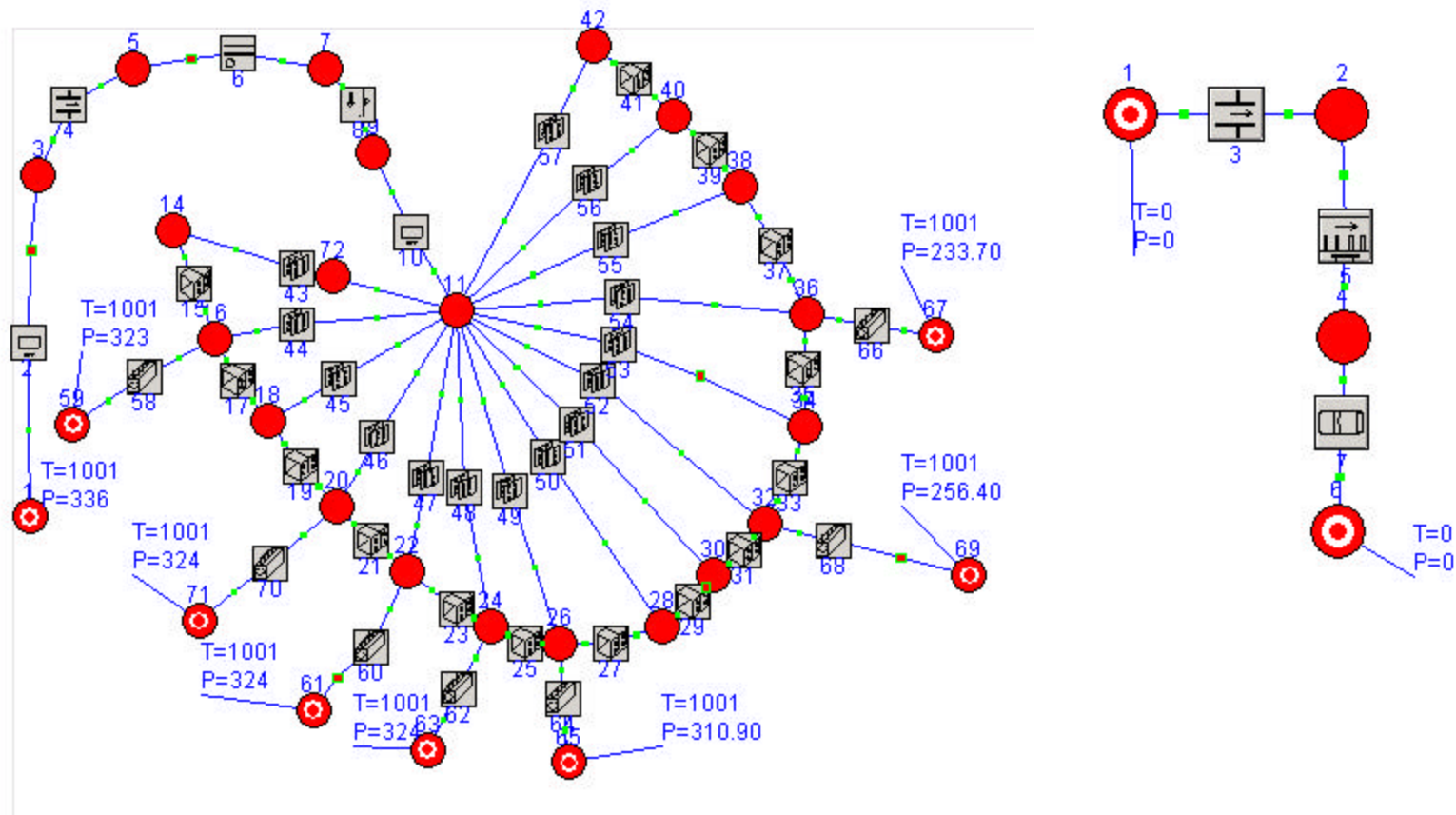
Thermal Modeling Methodology

- ◆ UATDMT: Thermal Design Management Tool
- ◆ UACTM: Compact Thermal Modeling Methodology for Electronic Packages
- ◆ UACTFHS: Compact Thermofluid Modeling Methodology for Heat Sinks



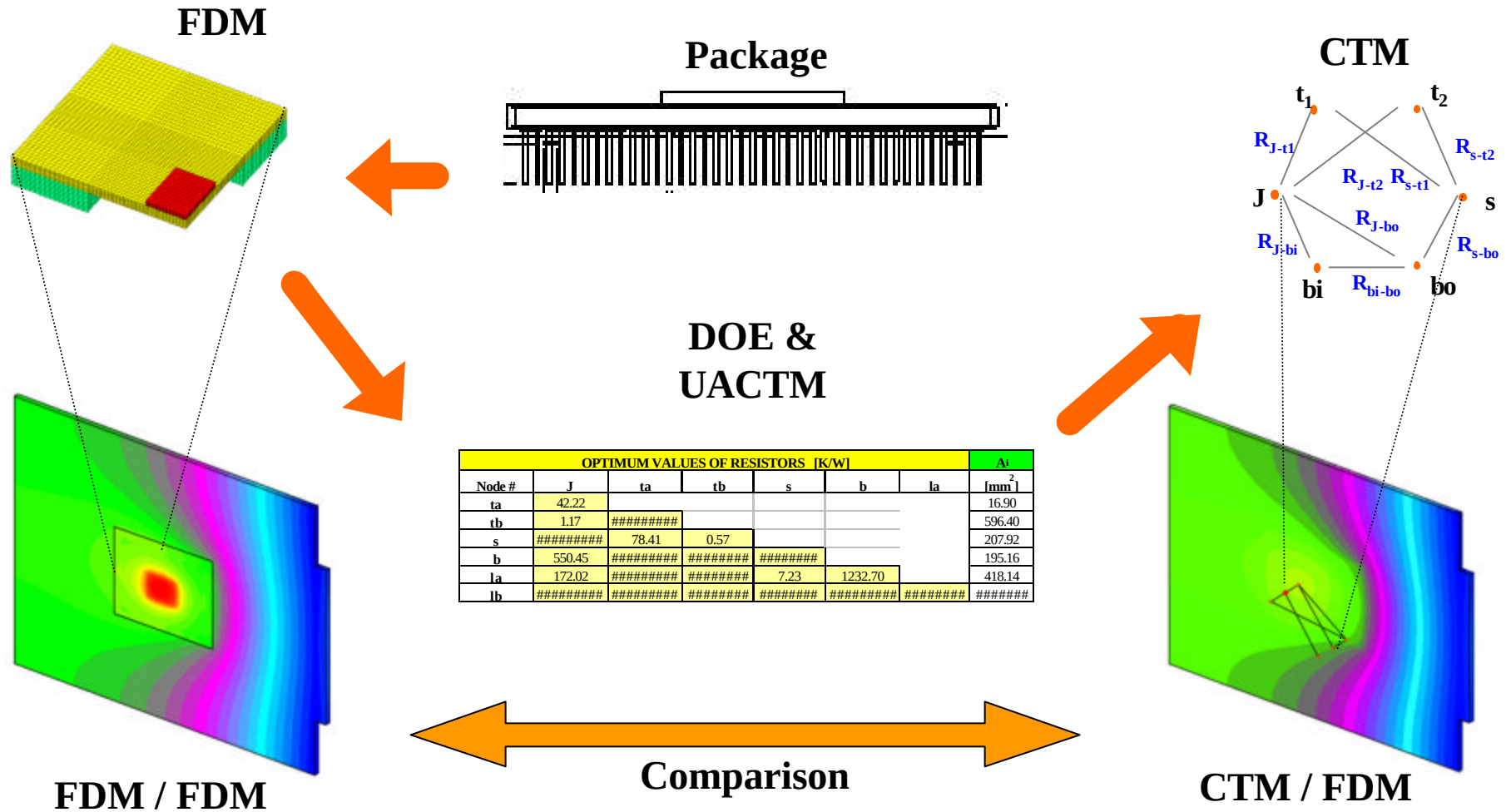


Models can be Simple or Complex



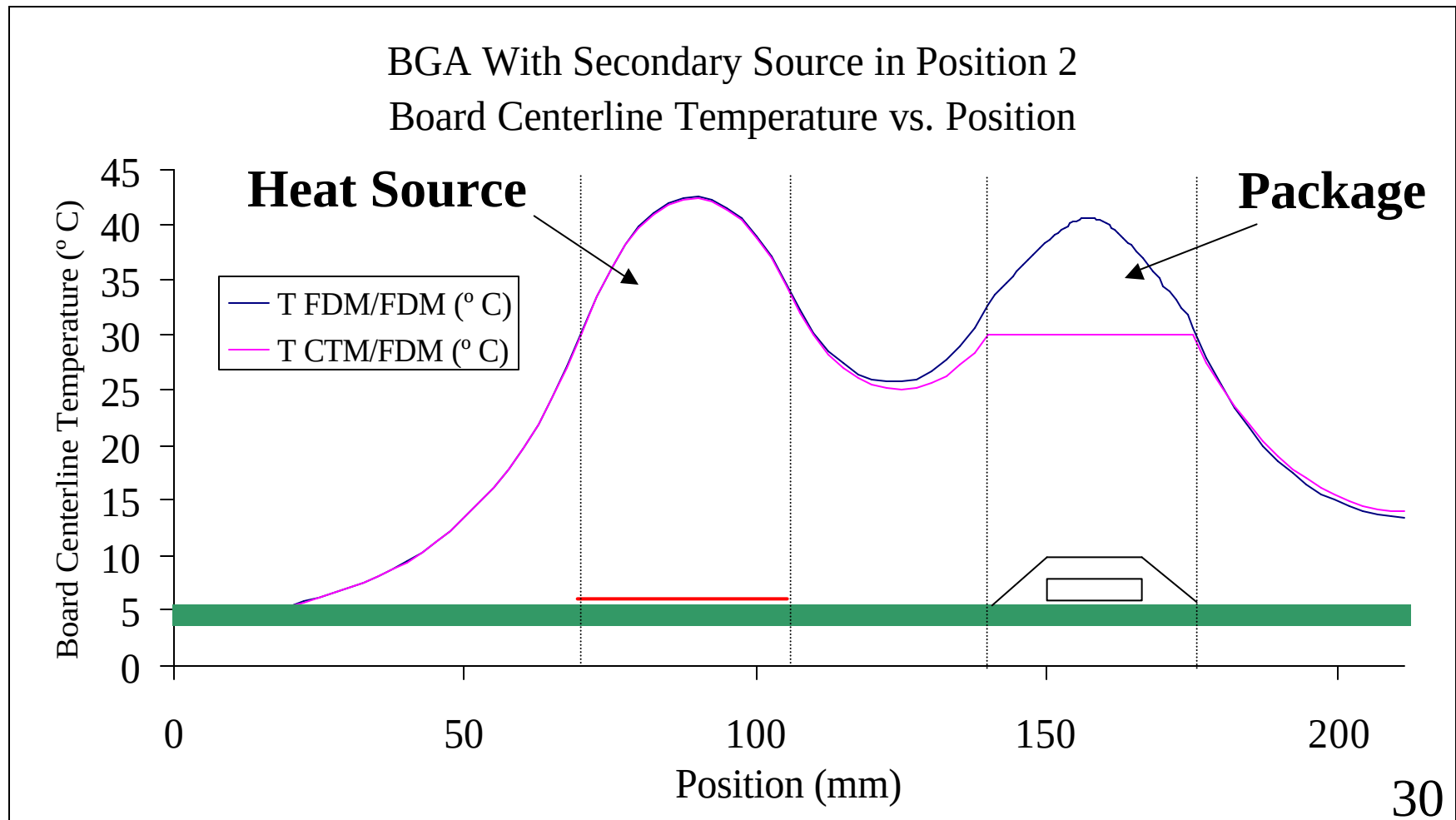


Compact Thermal Models of Packages





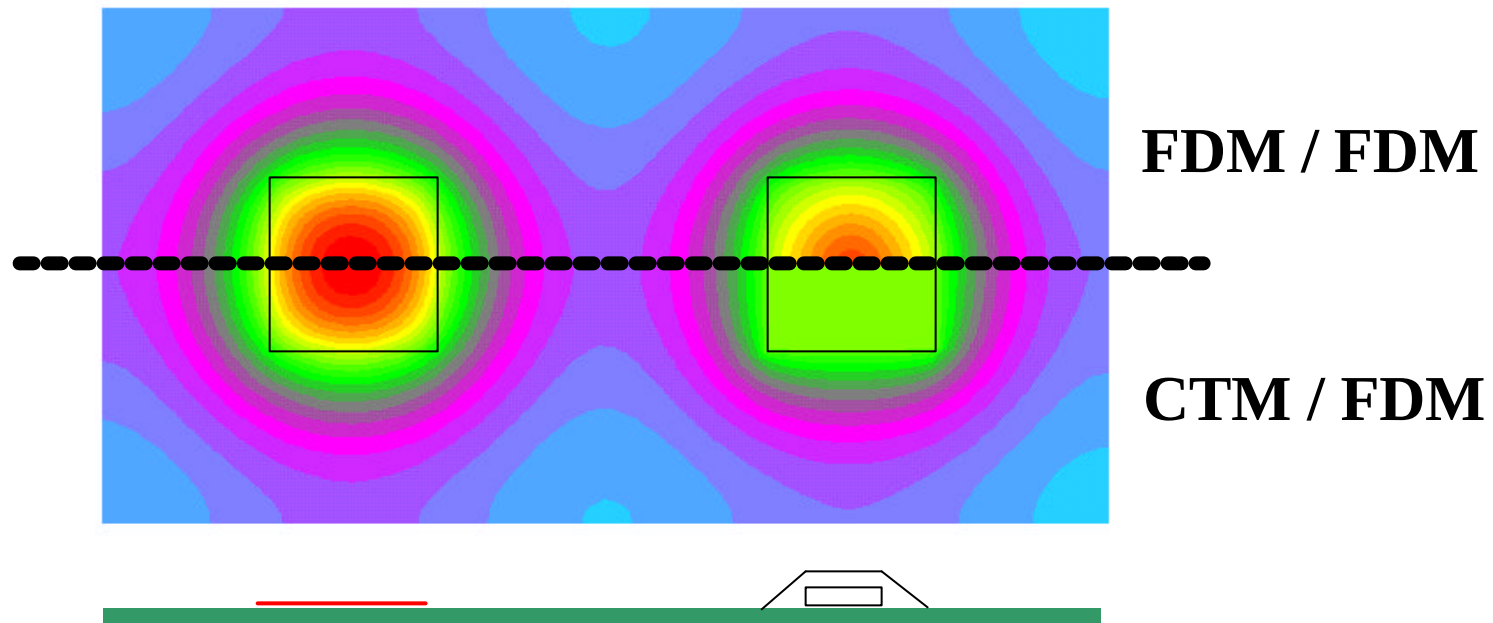
Results with Secondary Source





Results with Secondary Source

BGA Package: Position 3 Moderate Conductivity Board





Rapid Thermofluid Prototyping

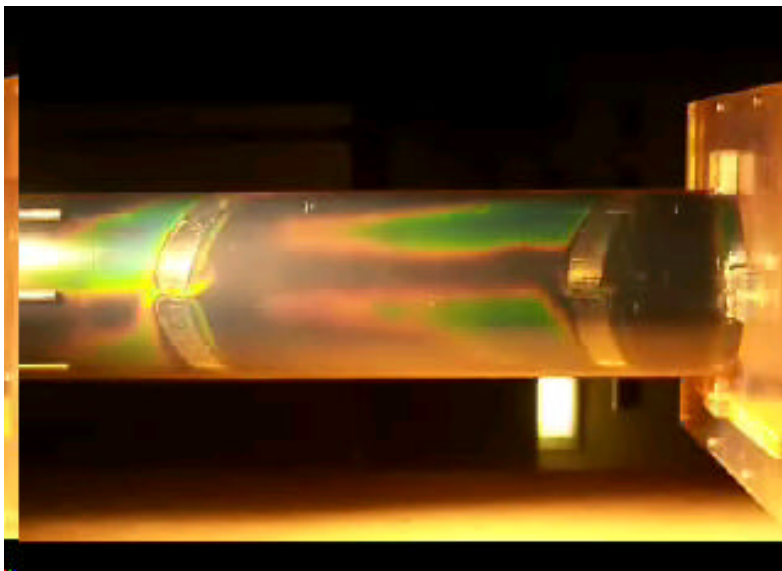
- ◆ Rapidly create testable prototype using Rapid Prototyping
 - Stereo Lithography especially promising
- ◆ Integrated total and static pressure tapping for hydraulic measurements
- ◆ Novel Transient methods for measuring local heat transfer



Rapid Thermofluid Prototyping



- SLA model of turbulated passage
- Integrated wall static taps
- Digital Transient Liquid Crystal technique for measuring htc
- Excellent spatial resolution





Recent References

1. E. Veksler, R. Loutfy, A. Ortega, D. Wallinger, and T. Prindiville, "Novel Boiling Enhanced Multi-Channel Heat Sinks," Paper TED-AJ03-380, 6th ASME-JSME Thermal Engineering Joint Conference, March 16-20, 2003.
2. Dogruoz, M. B., Urdaneta, M., Ortega, A., and Westphal, R. V., 2002, "[Experiments and Modeling of the Hydraulic Resistance of In-Line Square Pin Fin Heat Sinks With Top Bypass Flow](#)," Proc., ITherm Eighth Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, San Diego, California, Paper # 2002-1296.
3. Dogruoz, M. B., Urdaneta, M., and Ortega, A., 2002, "[Experiments and Modeling of the Heat Transfer of In-Line Square Pin Fin Heat Sinks With Top Bypass Flow](#)," Proc., IMECE2002 ASME International Mechanical Engineering Congress & Exposition, New Orleans, Louisiana, Paper # 2002-34245
4. Issa, J. and Ortega, A. "Experimental Measurements of the Flow and Heat Transfer of a Square Jet Impinging on an Array of Square Pin Fins," Procs. IMECE2002 ASME International Engineering Congress & Exposition, November 17-22, 2002, New Orleans, Louisiana, Paper # 2002-34244
5. Devoe, J. and Ortega, A. "An Assessment of the Behavior of Compact Thermal Models of Electronic Packages In a Printed Circuit Board Level Environment," Procs. 2002 ITherm Symposium, San Diego, May 2002.
6. Devoe, J. and Ortega, A. "An Investigation of Board Level Effects on Compact Thermal Models of Electronic Chip Packages," Procs. 2002 SEMITHERM Symposium, San Jose, March 2002.